

Que-1	Research and provide three real-world applications where C programming is extensively used, such as in embedded systems, operating systems, or game development.
Ans	Real-World Applications of C Programming:- C programming is widely used in many real-world applications because it provides: • High performance • Fast execution • Low-level memory access • Hardware control Even today, C remains one of the most important programming languages in technology and software development. Major Real-World Applications of C 1. Embedded Systems 2. Operating Systems 3. Game Development 1. Embedded Systems Embedded systems are specialized computer systems designed to perform specific tasks. C is heavily used in embedded systems because it: • Directly interacts with hardware • Uses less memory • Executes very fast Examples of Embedded Systems • Washing machines • Microwave ovens • Smart TVs • Traffic lights • Medical devices • Car control systems

Why C is Used in Embedded Systems

1. Fast execution
2. Efficient memory usage
3. Direct hardware access
4. Suitable for real-time systems

Real-World Example

Modern cars use embedded systems written in C for:

- Airbags
- Braking systems
- Engine control
- Safety sensors ([NareshIT](#))

2. Operating Systems

An operating system manages:

- Memory
- Hardware
- Processes
- Files

C is one of the most important languages for operating system development.

Operating Systems Using C

- UNIX
- Linux
- Windows (partially)
- macOS (partially)

Why C is Used in Operating Systems

1. Close to hardware
2. High performance
3. Efficient memory management
4. System-level programming support

Real-World Example

The UNIX operating system was rewritten in C, which made C the standard language for system programming. Modern operating systems still use C in kernels and drivers. ([GeeksforGeeks](#))

3. Game Development

C is also used in game development because games require:

- High speed
- Fast graphics processing
- Efficient memory usage

Uses in Game Development

- Game engines
- Graphics systems
- Audio systems
- Physics engines

Why C is Used in Games

1. High performance
2. Fast execution speed
3. Better hardware control
4. Efficient resource management

Real-World Example

Many classic games and game engines were developed using C and C-based technologies, including simple games like Snake and Tic-Tac-Toe. ([LearnTube](#))

Other Applications of C

1. Database systems
2. Compilers
3. Networking software
4. Device drivers

	5. IoT systems 6. Robotics 7. Medical equipment Why C is Still Popular Today 1. Fast and efficient 2. Portable across systems 3. Strong hardware interaction 4. Foundation for many modern languages 5. Widely used in industry
Que-2	Install a C compiler on your system and configure the IDE. Write your first program to print "Hello, World!" and run it.
Ans	<pre>#include<stdio.h> main(){ printf("Hello World....."); }</pre>
Que-3	Write a C program that includes variables, constants, and comments. Declare and use different data types (int, char, float) and display their values.
Ans	<pre>#include<stdio.h> #define PI 3.14 main(){ int a=10; char b='h'; float c=1.23; printf("\nValue of Integer is %d.",a); printf("\nValue of Char is %c.",b); printf("\nValue of Float is %f.",c); printf("\nValue of PI is %f.",PI); }</pre>
Que-4	Write a C program that accepts two integers from the user and performs arithmetic, relational, and logical operations on them. Display the results.

Ans	<pre> #include<stdio.h> main(){ int a,b; printf("Enter A Value:"); scanf("%d",&a); printf("Enter B Value:"); scanf("%d",&b); printf("\nArthmatic Operator"); printf("Addition is %d.\n",a+b); printf("Subtraction is %d.\n",a-b); printf("Mutliplication is %d.\n",a*b); printf("Divison is %d.\n",a/b); printf("\nRelational Operator\n"); printf("a>b=%d\n",a>b); printf("a<b=%d\n",a<b); printf("a==b=%d\n",a==b); printf("a!=b=%d\n",a!=b); printf("\nLogical Operator\n"); printf("(a>b && b>0)=%d\n",(a>b && b>0)); printf("(a>b b>0)=%d\n",(a>b b>0)); printf("!(a>b)=%d\n",!(a>b)); } </pre>
Que-5	Write a C program to check if a number is even or odd using an if-else statement. Extend the program using a switch statement to display the month name based on the user's input (1 for January, 2 for February, etc.).
Ans	<pre> #include<stdio.h> main(){ int num,month; </pre>

```
printf("Enter a number:");
scanf("%d",&num);

if(num%2==0){
    printf("%d is Even.\n",num);
}else{
    printf("%d is odd.\n",num);
}

printf("Enter number between 1 to 12::");
scanf("%d",&month);
switch(month){
    case 1:
        printf("This is January...");
        break;
    case 2:
        printf("This is Febuary...");
        break;
    case 3:
        printf("This is March...");
        break;
    case 4:
        printf("This is April...");
        break;
    case 5:
        printf("This is May...");
        break;
    case 6:
        printf("This is June...");
        break;
    case 7:
        printf("This is Julay...");
```

	<pre> break; case 8: printf("This is August..."); break; case 9: printf("This is Septembe..."); break; case 10: printf("This is October..."); break; case 11: printf("This is November..."); break; case 12: printf("This is December..."); break; default: printf("Invalid Month Number..."); break; } } </pre>
Que-6	Write a C program to print numbers from 1 to 10 using all three types of loops (while, for, do-while).
Ans	<pre> #include<stdio.h> main(){ int i; printf("\nFor Loop\n"); for(i=1;i<=10;i++){ printf("i=%d\n",i); } printf("\nWhile Loop\n"); i=1; </pre>

	<pre> while(i<=10){ printf("i=%d\n",i); i++; } printf("\nDo..While Loop\n"); i=1; do{ printf("i=%d\n",i); i++; }while(i<=10); } </pre>
Que-7	Write a C program that uses the break statement to stop printing numbers when it reaches 5. Modify the program to skip printing the number 3 using the continue statement.
Ans	<pre> #include<stdio.h> main(){ int i; printf("\nFor Loop\n"); for(i=1;i<=10;i++){ if(i==3){ continue; }else if(i==6){ break; } printf("i=%d\n",i); } } </pre>
Que-8	Write a C program that calculates the factorial of a number using a function. Include function declaration, definition, and call.
Ans	<pre> #include<stdio.h> void FunFact(); void FunFact(){ int num,i,fact=1; </pre>

	<pre> printf("Enter a Number::"); scanf("%d",&num); for(i=1;i<=num;i++){ fact*=i; } printf("Factorial of %d is %d.",num,fact); } int main(){ FunFact(); return 0; } </pre>
Que-9	Write a C program that stores 5 integers in a one-dimensional array and prints them. Extend this to handle a two-dimensional array (3x3 matrix) and calculate the sum of all elements.
Ans	<pre> #include<stdio.h> main(){ int a[99],i,j,num[99][99],sum; printf("\nOne-Dimensional Array\n"); for(i=1;i<=5;i++){ printf("Enter a number:"); scanf("%d",&a[i]); } for(i=1;i<=5;i++){ printf("Number is %d.\n",a[i]); } printf("\nTwo-Dimensional Array\n"); for(i=0;i<3;i++){ for(j=0;j<3;j++){ printf("Enter a number:"); scanf("%d",&num[i][j]); } } } </pre>

	<pre> for(i=0;i<3;i++){ for(j=0;j<3;j++){ sum+=num[i][j]; } printf("\n"); } printf("Sum of all Element is %d.",sum); } </pre>
Que-10	Write a C program to demonstrate pointer usage. Use a pointer to modify the value of a variable and print the result.
Ans	<pre> #include<stdio.h> main(){ int num=10; int *ptr; ptr=&num; printf("Original value is %d.\n",num); *ptr=50; printf("Modified value is %d.\n",num); } </pre>
Que-11	Write a C program that takes two strings from the user and concatenates them using strcat(). Display the concatenated string and its length using strlen().
Ans	<pre> #include<stdio.h> #include<string.h> main(){ char str1[99],str2[99],str3[99]; printf("Enter String 1:"); scanf("%s",str1); printf("Enter String 2:"); </pre>

	<pre>scanf("%s",str2); strcat(str1,str2); printf("Concatenated String is %s.\n",str1); printf("Lenght of String is %lu.",strlen(str1)); }</pre>
Que-12	Write a C program that defines a structure to store a student's details (name, roll number, and marks). Use an array of structures to store details of 3 students and print them.
Ans	<pre>#include<stdio.h> struct Student{ int rollNo; char name[99]; int marks; }; main(){ struct Student s[3]; int i; for(i=0;i<3;i++){ printf("Enter details of Student's %d\n.",i+1); printf("\nEnter Roll No:-"); scanf("%d",&s[i].rollNo); printf("\nEnter a Name:"); scanf("%s",&s[i].name); printf("\nEnter Marks:"); scanf("%d",&s[i].marks); } printf("Student's Details\n"); for(i=0;i<3;i++){ printf("Student %d\n",i+1); printf("Student Roll no is %d.\t",s[i].rollNo); printf("Student Name is %s.\t",s[i].name); } }</pre>

	<pre>printf("Student Mark is %d.\n",s[i].marks); } }</pre>
Que-13	Write a C program to create a file, write a string into it, close the file, then open the file again to read and display its contents.
Ans	<pre>#include<stdio.h> int main(){ FILE *fp; char text[100]; fp=fopen("data.txt","w"); fprintf(fp,"Hello File Handling"); fclose(fp); fp=fopen("data.txt","r"); fgets(text,100,fp); printf("File Content:%s",text); fclose(fp); return 0; }</pre>
Extra Lab Excercise for Logic Building	
Que-1	Write a C program that acts as a simple calculator. The program should take two numbers and an operator as input from the user and perform the respective operation (addition, subtraction, multiplication, division, or modulus) using operators. Challenge: Extend the program to handle invalid operator inputs.
Ans	<pre>#include<stdio.h> main(){ int num1,num2; char op; printf("\nEnter number 1:"); scanf("%d",&num1); printf("\nEnter Operator(+,-,*,/,%):"); scanf(" %c",&op);</pre>

	<pre> printf("\nEnter number 2:"); scanf("%d",&num2); switch(op){ case '+': printf("Addition is %d.",num1+num2); break; case '-': printf("Subtraction is %d.",num1-num2); break; case '*': printf("Multiplication is %d.",num1*num2); break; case '/': printf("Division is %d.",num1/num2); break; case '%': printf("Module is %d.",num1%num2); break; default: printf("Invalid Oprator.."); } } </pre>
Que-2	Write a C program that takes an integer from the user and checks the following using different operators: - o Whether the number is even or odd. - o Whether the number is positive, negative, or zero. - o Whether the number is a multiple of both 3 and 5.
Ans	<pre> #include<stdio.h> main(){ int num; printf("Enter a number:"); scanf("%d",&num); </pre>

	<pre> if(num%2==0){ printf("%d is even.\n",num); }else{ printf("%d is Odd.\n",num); } if(num>0){ printf("\n%d is Positive.",num); }else if(num<0){ printf("\n%d is Negetive.",num); }else{ printf("\n%d is Zero.",num); } if(num%3==0 && num%5==0){ printf("\n%d is Multiplty by both 3 And 5."); }else{ printf("\n%d is not Multiplty by both 3 And 5."); } } </pre>
Que-3	Write a C program that takes the marks of a student as input and displays the corresponding grade based on the following conditions: o Marks > 90: Grade A o Marks > 75 and <= 90: Grade B o Marks > 50 and <= 75: Grade C o Marks <= 50: Grade D • Use if-else or switch statements for the decision-making process.
Ans	<pre> #include<stdio.h> main(){ int mark; printf("Enter a student mark:"); scanf("%d",&mark); if(mark>90){ printf("Grade A..."); }else if(mark>75 && mark<=90){ </pre>

	<pre> printf("Grade B..."); }else if(mark>50 && mark<=75){ printf("Grade C..."); }else if(mark<=50){ printf("Grade D..."); } } </pre>
Que-4	Write a C program that takes three numbers from the user and determines: o The largest number. o The smallest number. Challenge: Solve the problem using both if-else and switch-case statements.
Ans	<pre> #include<stdio.h> main(){ int num1,num2,num3,largest,smallest; printf("Enter a number 1:"); scanf("%d",&num1); printf("Enter a number 2:"); scanf("%d",&num2); printf("Enter a number 3:"); scanf("%d",&num3); printf("Find with IF...else Statement....\n"); if(num1>num2 && num1>num3){ largest=num1; }else if(num2>num1 && num2>num3){ largest=num2; }else{ largest=num3; } if(num1<num2 && num1<num3){ smallest=num1; }else if(num2<num1 && num2<num3){ </pre>

```
        smallest=num2;

    }else{
        smallest=num3;
    }

    printf("\nLargest Number is %d.\n",largest);
    printf("\nSmallest Number is %d.\n",smallest);

    printf("Find with Switch Statement....\n");

    if(num1>num2 && num1>num3){
        largest=1;
    }else if(num2>num1 && num2>num3){
        largest=2;
    }else{
        largest=3;
    }

    switch(largest){
        case 1:
            printf("%d is the Largest....\n",num1);
            break;
        case 2:
            printf("%d is the Largest....\n",num2);
            break;
        case 3:
            printf("%d is the Largest....\n",num3);
            break;
    }

    if(num1<num2 && num1<num3){
        smallest=1;
```


	<pre> }else if(num2<num1 && num2<num3){ smallest=2; }else{ smallest=3; } switch(smallest){ case 1: printf("%d is the smallest....\n",num1); break; case 2: printf("%d is the smallest....\n",num2); break; case 3: printf("%d is the smallest....",num3); break; } } } </pre>
Que-5	Write a C program that checks whether a given number is a prime number or not using a for loop. Challenge: Modify the program to print all prime numbers between 1 and a given number.
Ans	<pre> #include<stdio.h> main(){ int num,i,j,count; printf("Enter a number:"); scanf("%d",&num); for(i=1;i<=num;i++){ count=0; for(j=1;j<=i;j++){ if(i%j==0){ count++; } } } } </pre>

	<pre> } } if(count==2){ printf("%d ",i); } } } </pre>
Que-6	Write a C program that takes an integer input from the user and prints its multiplication table using a for loop. Challenge: Allow the user to input the range of the multiplication table (e.g., from 1 to N).
Ans	<pre> #include<stdio.h> main(){ int num,i,j; printf("Enter a Number of Range(1-N):"); scanf("%d",&num); for(i=1;i<=10;i++){ for(j=1;j<=num;j++){ printf("%d * %d =%d\t",j,i,j*i); } printf("\n"); } } </pre>
Que-7	Write a C program that takes an integer from the user and calculates the sum of its digits using a while loop. Challenge: Extend the program to reverse the digits of the number.
Ans	<pre> #include<stdio.h> main(){ int num,rev=0,sum=0,rem,temp; printf("Enter a number:"); scanf("%d",&num); temp=num; printf("\nSum Of Digit....."); </pre>

	<pre> while(num>0){ rem=num%10; sum+=rem; num=num/10; } printf("\nSum of Digit is %d.",sum); printf("\nRevers Number....."); while(temp!=0){ rem=temp%10; rev=rev*10+rem; temp=temp/10; } printf("\nReverse Number is %d.",rev); } </pre>
Que-8	Write a C program that accepts 10 integers from the user and stores them in an array. The program should then find and print the maximum and minimum values in the array. Challenge: Extend the program to sort the array in ascending order.
Ans	<pre> #include<stdio.h> int main(){ int num[10],max,min,sort; int i,j; printf("Enter 10 integer number\n"); for(i=0;i<10;i++){ scanf("%d",&num[i]); } max=num[0]; min=num[0]; printf("\nFind Maximum value in Array.....\n"); for(i=1;i<10;i++){ </pre>

	<pre> if(num[i]>max){ max=num[i]; } } printf("Largest Element in array is %d.",max); printf("\nFind Minimum value in Array.....\n"); for(i=1;i<10;i++){ if(num[i]<min){ min=num[i]; } } printf("\nMinimum Element in array is %d.",min); printf("\nSort Array in Accending order.....\n"); for(i=0;i<10;i++){ for(j=i+1;j<10;j++){ if(num[i]>num[j]){ sort=num[i]; num[i]=num[j]; num[j]=sort; } } } for(i=0;i<10;i++){ printf("%d ",num[i]); } } </pre>
Que-9	Write a C program that accepts two 2x2 matrices from the user and adds them. Display the resultant matrix. Challenge: Extend the program to work with 3x3 matrices and matrix multiplication.

Ans	<pre>#include<stdio.h> main(){ int a1[2][2],b1[2][2],c1[2][2],i,j; int a[3][3],b[3][3],c[3][3],r,w,k; printf("2x2 Resulant Matrix.....\n"); printf("\n"); printf("Enter Matrix A element\n"); for(i=0;i<2;i++){ for(j=0;j<2;j++){ scanf("%d",&a1[i][j]); } } printf("Enter Matrix B element\n"); for(i=0;i<2;i++){ for(j=0;j<2;j++){ scanf("%d",&b1[i][j]); } } printf("Addition of Matrix Element\n"); for(i=0;i<2;i++){ for(j=0;j<2;j++){ c1[i][j]=a1[i][j]+b1[i][j]; } } for(i=0;i<2;i++){ for(j=0;j<2;j++){ printf("%d ",c1[i][j]); } printf("\n"); } printf("\n");</pre>
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	<pre> printf("3x3 Matrix.....\n"); printf("\n"); printf("Enter Matrix A element\n"); for(r=0;r<3;r++){ for(w=0;w<3;w++){ scanf("%d",&a[r][w]); } } printf("Enter Matrix B element\n"); for(r=0;r<3;r++){ for(w=0;w<3;w++){ scanf("%d",&b[r][w]); } } printf("Multiplication Of Matrix\n"); for(r=0;r<3;r++){ for(w=0;w<3;w++){ c[r][w]=0; for(k=0;k<3;k++){ c[r][w]+=a[r][k]*b[k][w]; } } } for(r=0;r<3;r++){ for(w=0;w<3;w++){ printf("%d ",c[r][w]); } printf("\n"); } } </pre>
Que-10	Write a C program that takes N numbers from the user and stores them in an array. The program should then calculate and display the sum of all array

	elements. Challenge: Modify the program to also find the average of the numbers.
Ans	<pre>#include<stdio.h> main(){ int a[99],sum,i,n; float avg; printf("How Many Element you want in Array:"); scanf("%d",&n); printf("\nInteger Array\n"); printf("Enter Array Element's:\n",i); for(i=1;i<=n;i++){ scanf("%d",&a[i]); sum+=a[i]; } printf("\nAddition of Array Element is %d",sum); printf("\n"); printf("Average of Array Element's\n"); for(i=1;i<=n;i++){ avg=sum/n; } printf("\nAverage of Array Element is %f",avg); }</pre>
Que-11	Write a C program that generates the Fibonacci sequence up to N terms using a recursive function. • Challenge: Modify the program to calculate the Nth Fibonacci number using both iterative and recursive methods. Compare their efficiency.
Ans	<pre>#include<stdio.h> int recFib(int n){ if(n==0){ return 0; } if(n==1){</pre>

```
        return 1;
    }
    return recFib(n-1)+recFib(n-2);
}
int itrFib(int n){
    int a=0,b=1,c,i;
    if(n==0){
        return 0;
    }
    for(i=2;i<=n;i++){
        c=a+b;
        a=b;
        b=c;
    }
    return b;
}
main(){
    int n,i,recAns,itrAns;
    printf("\nFibonacci Series using Recursion Function.....\n");
    printf("Enter a number:");
    scanf("%d",&n);
    for(i=0;i<n;i++){
        printf("%d ",recFib(i));
    }
    printf("\nFibonacci Series using Iterative Method.....\n");
    printf("Enter a number:");
    scanf("%d",&n);
    recAns=recFib(n);
    itrAns=itrFib(n);
    printf("Fibonacci series using recursion=%d\n",recAns);
    printf("Fibonacci series using iteration=%d\n",itrAns);
}
```


Que-12	Write a C program that calculates the factorial of a given number using a function. Challenge: Implement both an iterative and a recursive version of the factorial function and compare their performance for large numbers.
Ans	<pre>#include<stdio.h> long recursiveFact(int n){ if(n==0 n==1){ return 1; } return n*recursiveFact(n-1); } long iterativeFact(int n){ long fact=1; int i; for(i=1;i<=n;i++){ fact*=i; } return fact; } int main(){ int num; printf("Enter a Number: "); scanf("%d",&num); printf("\nIterative Factorial = %ld",iterativeFact(num)); printf("\nRecursive Factorial = %ld",recursiveFact(num)); return 0; }</pre>
Que-13	Write a C program that takes a number as input and checks whether it is a palindrome using a function. Challenge: Modify the program to check if a given string is a palindrome.
Ans	<pre>#include<stdio.h> #include<string.h> int numPalidrom(){</pre>

```
int num,rev=0,temp,rem;

printf("Enter a number:");
scanf("%d",&num);

temp=num;

while(temp!=0){
    rem=temp%10;
    rev=rev*10+rem;
    temp/=10;
}
if(num==rev){
    printf("%d is Palindrome Number...\n",num);
}else{
    printf("%d is not Palindrome Number...\n",num);
}
}

int strPalindrome(){
    char str[99];
    int i,len,flag=1;

    printf("Enter a String:");
    scanf("%s",str);
    len=strlen(str);
    for(i=0;i<len/2;i++){
        if(str[i]!=str[len-1-i]){
            flag=0;
            break;
        }
    }
    if(flag==1){
```

	<pre> printf("%s is palindromi number...\n",str); }else{ printf("%s is not palindromi number...\n",str); } } main(){ numPalidrom(); strPalindrome(); } </pre>
Que-14	Write a C program that takes a string as input and reverses it using a function. Challenge: Write the program without using built-in string handling functions.
Ans	<pre> #include<stdio.h> int revString(){ char str[99]; int i,len=0; printf("Enter A String:"); scanf("%s",str); while(str[len]!='\0'){ len++; } printf("\nRevered String.....\n"); for(i=len-1;i>=0;i--){ printf("%c",str[i]); } } main(){ revString(); } </pre>
Que-15	Write a C program that takes a string from the user and counts the number of vowels and consonants in the string.

	Challenge: Extend the program to also count digits and special characters.
Ans	<pre> #include<stdio.h> int main(){ char str[100]; int i=0,vowel=0,consonant=0,digit=0,special=0; printf("Enter a String: "); scanf("%s",str); while(str[i]!='\0'){ if(str[i]=='a' str[i]=='e' str[i]=='i' str[i]=='o' str[i]=='u' str[i]=='A' str[i]=='E' str[i]=='I' str[i]=='O' str[i]=='U'){ vowel++; } else if((str[i]>='a' && str[i]<='z') (str[i]>='A' && str[i]<='Z')){ consonant++; } else if(str[i]>='0' && str[i]<='9'){ digit++; } else{ special++; } i++; } printf("\nVowels = %d",vowel); printf("\nConsonants = %d",consonant); printf("\nDigits = %d",digit); printf("\nSpecial Characters = %d",special); return 0; } </pre>
Que-16	Write a C program that counts the number of words in a sentence entered by the user. • Challenge: Modify the program to find the longest word in the sentence.

Ans	<pre>#include<stdio.h> #include<string.h> int main(){ char str[200],longest[50]; int i,word=1,length=0,maxLength=0,start=0; printf("Enter a Sentence: "); gets(str); for(i=0;str[i]!='\0';i++){ if(str[i]==' '){ word++; } if(str[i]!=' ' && str[i]!='\0'){ length++; } else{ if(length>maxLength){ maxLength=length; strncpy(longest,&str[start],length); longest[length]='\0'; } length=0; start=i+1; } } if(length>maxLength){ maxLength=length; strncpy(longest,&str[start],length); longest[length]='\0'; } printf("\nTotal Words = %d",word); printf("\nLongest Word = %s",longest); return 0;</pre>
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	}
Extra Logic Buileding Challenges	
Que-1	Write a C program that checks whether a given number is an Armstrong number or not (e.g., $153 = 1^3 + 5^3 + 3^3$). Challenge: Write a program to find all Armstrong numbers between 1 and 1000.
Ans	<pre>#include<stdio.h> int main(){ int num,temp,rem,sum=0,i; printf("Enter a Number: "); scanf("%d",&num); temp=num; while(temp!=0){ rem=temp%10; sum=sum+(rem*rem*rem); temp=temp/10; } if(sum==num){ printf("%d is Armstrong Number\n",num); } else{ printf("%d is Not Armstrong Number\n",num); } printf("\nArmstrong Numbers Between 1 and 1000:\n"); for(i=1;i<=1000;i++){ temp=i; sum=0; while(temp!=0){ rem=temp%10; sum=sum+(rem*rem*rem); temp=temp/10; } } }</pre>

	<pre> } if(sum==i){ printf("%d ",i); } } return 0; } </pre>
Que-2	Write a C program that generates Pascal's Triangle up to N rows using loops. Challenge: Implement the same program using a recursive function.
Ans	<pre> #include<stdio.h> int pascal(int n,int r){ if(r==0 r==n){ return 1; } return pascal(n-1,r-1)+pascal(n-1,r); } int main(){ int n,i,j; printf("Enter Number of Rows: "); scanf("%d",&n); for(i=0;i<n;i++){ for(j=0;j<=i;j++){ printf("%d ",pascal(i,j)); } printf("\n"); } return 0; } </pre>
Que-3	Write a C program that implements a simple number guessing game. The program should generate a random number between 1 and 100, and the user should guess the number within a limited number of attempts. Challenge: Provide hints to the user if the guessed number is too high or too low.

Ans	<pre>#include<stdio.h> #include<stdlib.h> #include<time.h> int main(){ int number,guess,i; srand(time(0)); number=rand()%100+1; printf("Guess the Number Between 1 and 100\n"); printf("You Have 5 Attempts\n"); for(i=1;i<=5;i++){ printf("\nAttempt %d: ",i); scanf("%d",&guess); if(guess==number){ printf("Correct Guess! You Win!\n"); break; } else if(guess>number){ printf("Too High!\n"); } else{ printf("Too Low!\n"); } } if(guess!=number){ printf("\nYou Lost!"); printf("\nCorrect Number Was %d",number); } return 0; }</pre>
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